6. (a) Find the general solution of the differential equation, $x^3y'' - 2xy = 6\ln x;\ x > 0$. [4]

(b) Find the general solution of the following equation by the method of variation of parameters. [3]

$$y'' - 5y' + 6y = 2e^{x}.$$

7. (a) Form a partial differential equation from $z(x, y) = y^2 + 2f(\frac{1}{x} + \ln y)$, where f is an arbitrary function. $x > 0$. and $y > 0$. [3]

(b) Solve the partial differential equation $x^2\dfrac{\partial z}{\partial x} + y^2\dfrac{\partial z}{\partial y} = (x + y)z$. [5]